

Haoqi Yuan

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EDUCATION

Peking University - School of Computer Science: *Ph.D. Candidate* 2021.9 - Present

- Supervisor: Professor Zongqing Lu
- Research Area: Reinforcement Learning, Embodied AI
- Twice awarded the Peking University Presidential Scholarship.

Peking University - Turing Class: *Bachelor's Degree* 2017.9 - 2021.7

- Major: Computer Science and Technology, GPA 3.79/4, rank 14th/232
- Received National Scholarship and Peking University Merit Student Pacesetter.

INTERNSHIPS

BeingBeyond 2025 - Present

- Study scalable learning methods for dexterous manipulation and humanoid robots.

Beijing Academy of Artificial Intelligence (BAAI) 2023 - 2025

- Study efficient open-world agents equipped with RL and Foundation Models.
- Study RL for dexterous manipulation.

RECENT WORKS

Haoqi Yuan, Ziyue Huang, Ye Wang, Chuan Mao, Chaoyi Xu, Zongqing Lu. DemoGrasp: Universal Dexterous Grasping from a Single Demonstration, 2025 (arXiv:2509.22149).

- An efficient RL framework for universal dexterous grasping, achieving high success rate from simulation to the real world for any dexterous hand.

Hao Luo, Yicheng Feng, Wanpeng Zhang, Sipeng Zheng, Ye Wang, **Haoqi Yuan**, Jiazheng Liu, Chaoyi Xu, Qin Jin, Zongqing Lu. Being-H0: Vision-Language-Action Pretraining from Large-Scale Human Videos, 2025 (arXiv:2507.15597).

- The first VLA foundation model trained on large-scale human videos and hand motions, efficiently transferring to robotic dexterous manipulation tasks.

Haoqi Yuan, Yu Bai, Yuhui Fu, Bohan Zhou, Yicheng Feng, Xinrun Xu, Yi Zhan, Börje F. Karlsson, Zongqing Lu. Being-0: A Humanoid Robotic Agent with Vision-Language Models and Modular Skills, 2025 (arXiv:2503.12533).

- We propose a hierarchical agent framework for humanoid robots, solving long-horizon, embodied tasks autonomously.

PUBLICATIONS

Haoqi Yuan, Bohan Zhou, Yuhui Fu, Zongqing Lu. Cross-Embodiment Dexterous Grasping with Reinforcement Learning, 2024.

- **First Author**, accepted at **ICLR 2025**.
- We propose a reinforcement learning method for training cross-embodiment dexterous grasping policies.

Haoqi Yuan, Yuhui Fu, Feiyang Xie, Zongqing Lu. Pre-Trained Multi-Goal Transformers with Prompt Optimization for Efficient Online Adaptation, 2024.

- **First Author**, accepted at **NeurIPS 2024**.
- We propose a pre-training and fine-tuning method for reinforcement learning that efficiently explores and adapts to long-horizon tasks.

Haoqi Yuan, Zhancun Mu, Feiyang Xie, Zongqing Lu. Pre-Training Goal-Based Models for Sample-Efficient Reinforcement Learning, 2024.

- **First Author**, accepted at **ICLR 2024**, selected for **oral presentation (top 1.2%)**
- We propose pre-training goal-based models to efficiently learn bi-level policies for solving long-horizon tasks in open-world environments.

Haoqi Yuan, Chi Zhang, Hongcheng Wang, Feiyang Xie, Penglin Cai, Hao Dong, Zongqing Lu. Plan4MC: Skill Reinforcement Learning and Planning for Open-World Minecraft Tasks, 2023.

- **First Author**, accepted at **NeurIPS 2023 Workshop**
- We develop an open-world agent that integrates LLM-assisted planning and reinforcement learning, solving 40 challenging tasks in Minecraft.

Haoqi Yuan, Zongqing Lu. Robust Task Representations for Offline Meta-Reinforcement Learning via Contrastive Learning, 2022.

- **First Author**, accepted at **ICML 2022**
- We proposed a task representation method combining contrastive learning and generative models to address distribution shift issues in offline meta-reinforcement learning.

Haoqi Yuan, Ruihai Wu, Andrew Zhao, Haipeng Zhang, Zihan Ding, Hao Dong. DMotion: Robotic Visuomotor Control with Unsupervised Forward Model Learned from Videos, 2021.

- **First Author**, accepted at **IROS 2021**
- We present a data-efficient approach for model-based control by learning motion representations and world models from unsupervised videos.

Ziye Huang, **Haoqi Yuan**, Yuhui Fu, Zongqing Lu. Efficient Residual Learning with Mixture-of-Experts for Universal Dexterous Grasping, 2024.

- Accepted at **ICLR 2025**
- We propose a universal dexterous grasping method based on residual policy learning and mixture-of-experts, achieving SOTA performance on DexGraspNet.

Shaoteng Liu, **Haoqi Yuan**, Minda Hu, Yanwei Li, Yukang Chen, Shu Liu, Zongqing Lu, Jiaya Jia. RL-GPT: Integrating Reinforcement Learning and Code-as-policy, 2024.

- Accepted at **NeurIPS 2024**, selected for **oral presentation**
- We propose using reinforcement learning as a tool for LLM-based agents. Our agent is the first to autonomously learn to mine diamonds in Minecraft without supervision.

Chi Zhang, Penglin Cai, Yuhui Fu, **Haoqi Yuan**, Zongqing Lu. Creative Agents: Empowering Agents with Imagination for Creative Tasks, 2025.

- Accepted at **UAI 2025**.
- We develop a framework and benchmark for creative agents, equipping them with imagination through generative models and LLMs.

AWARDS AND HONORS

Tun Scholarship	2025
Peking University Merit Student	2025
Luo Yuehua Scholarship	2024
Award for Scientific Research, Peking University	2024
Outstanding Student, National Engineering Research Center of Visual Technology	2022
Award for Scientific Research, Peking University	2022
Peking University Presidential Scholarship	2022
Peking University Presidential Scholarship	2021
John Hopcroft Scholarship	2020
Peking University Turing Class Scholarship	2019
National Scholarship	2018
Peking University Merit Student Pacesetter, Peking University	2018